



Brief article

Memory for pro-social intentions: When competing motives collide

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ABSTRACT

Memory for future actions, or *prospective memory* (PM), often involves remembering to do things for others. The present article explores the motivational mechanisms underlying memory for pro-social intentions through the manipulation of the social relevance of goals and presence of material rewards during an activity-based PM task. Results revealed that memory for the intention was better under pro-social than under standard conditions. However, when a material reward was introduced under pro-social conditions, it had the detrimental effect of reducing prospective remembering. Ongoing task performance was faster under pro-social than under No PM and standard PM conditions and it was unrelated to PM performance. In addition, the outcome of the ratings of two independent groups of participants confirmed that people are not aware of the potential conflict between pro-social and self-gain motives. Taken together, these new findings extend current PM theories by suggesting a prominent role of latent motivational mechanisms in guiding memory for pro-social intentions.

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1. Introduction

Remembering to perform a planned action in the future, or *prospective memory*, is a multifaceted phenomenon that can involve a wide variety of processes (Brandimonte, Einstein, & McDaniel, 1996; Kliegel, Einstein, & McDaniel, 2008). Yet, very little is known on the social and motivational determinants of prospective memory. In fact, in everyday life, memory for future actions is mostly socially driven and it often involves that people remember to do actions that benefit others rather than themselves (Brandimonte & Ferrante, 2008). In an early study, Meacham and Kushner (1980) reported that social tasks such as meeting someone were more likely to be performed than tasks of a less social nature. Since then, only a few studies involved some motivational (Cicogna & Nigro, 1998), social (Kvavilashvili, 1987), or importance-related (Kliegel, Martin, McDaniel, & Einstein, 2001; Kliegel, Martin, McDaniel, & Einstein, 2004) aspects of PM performance and, yet, those

studies were not primarily aimed at investigating the motivational mechanisms underlying prospective memory, or the social factors that can modulate motivation to remember. Thus, although motivation is generally recognized as a fundamental element of prospective memory, it seems to be under-explored.

A critical element of motivation is that it directs behavior towards specific goals and can be modulated by such factors as the presence of rewards. Typically, rewards work as positive incentives. They may increase motivation and this has a positive impact on cognitive performance (e.g., Pochon et al., 2002). However, providing a reward does not automatically and consistently instill motivation in participants. Much depends on the types and amount of reward, as well as on the way the reward is perceived by the person (Locke & Braver, 2008). For instance, when rewards are introduced for doing an intrinsically interesting activity (i.e., those that individuals would do in the absence of rewards, see Deci & Ryan, 2000), people tend to feel controlled by the rewards, hence decreasing their intrinsic motivation and, as a consequence, their behavior. A meta-analysis of 128 studies spanning three decades

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showed that not only monetary rewards, but also all contingent tangible rewards significantly undermined intrinsic motivation (Deci, Koestner, & Ryan, 1999). Recent findings in the economic literature revealed that receiving a small reward for a pro-social action results in a detrimental effect of *reducing* altruistic behavior (Bénabou & Tirole, 2006; Fehr & Rockenbach, 2003; Frey & Jegen, 2001). This effect, known as motivation *crowding out*, has been observed in a variety of social interactions both in field (Gneezy & Rustichini, 2000) and laboratory (Fehr & Rockenbach, 2003) experiments and it has been explained as a shift in the way people perceive the pro-social activity: Performing a pro-social activity without reward is clearly identifiable as intrinsically motivated altruism. However, once a reward is attached, people's intrinsic motivation decreases and so does altruistic behavior.

Interestingly, most recent research (Vohs, Mead, & Goode, 2006) using mental priming techniques showed that participants primed with money volunteered less time relative to control participants and donated significantly less money to a student fund than participants not primed with money. Thus, rewards may also prime unconscious motivational systems and this might be relevant to pro-social memory mechanisms.

We, therefore, addressed the issues of whether people's memory for delayed pro-social intentions is sensitive to the introduction of rewards and whether pro-social intention retrieval requires resources or it can be triggered and then operate outside of the person's awareness. To date, no current theory of PM has taken into consideration memory for pro-social intentions and its motivational underpinnings. This research is an initial attempt to fill this gap by investigating people's memory for actions that would benefit either the person herself or another person. More specifically, we explored whether activity-based prospective memory: (a) improves under pro-social conditions as compared to standard conditions; (b) is differentially influenced by the introduction of rewards according to the social nature of the task; (c) is associated with costs in the ongoing task performance. Additionally, we explored whether, in everyday life, people are aware of the potential detrimental effects of rewards.

2. Experiment

Theories of both cognitive and social psychology suggest that motivational states are characterized by enhanced accessibility of motivation-related concepts (Anderson, 1983; Bargh, Gollwitzer, Lee-Chai, Barndollar, & Troetschel, 2001; Bruner, 1957; Förster, Lieberman, & Higgins, 2005). Once activated, the goal's concept operates over extended time periods, hence guiding behavior towards the goal without the person's conscious monitoring (Gollwitzer & Bargh, 2005). Similarly, research on automaticity effects indicated that social behavior can be triggered and then operate in the absence of conscious guidance (Bargh & Morsella, 2008).

To explore whether motivational factors such as goals' social value and presence of rewards can have effects on prospective memory that are not mediated by strategic

processes, in the present study, an activity-based (Kvavilashvili & Ellis, 1996) prospective memory paradigm was used. Participants had to remember to do a simple action (signing a form) *at the end* of a highly demanding activity (verb identification). Doing one thing before or after another is the simplest situation among prospective memory tasks, because it does not require the interruption of an ongoing activity. In addition, in this experiment, the delay between the formation of the intention and the opportunity to realize the action was substantially longer than in typical laboratory-based prospective memory tasks (11 min vs. 2–4 min). According to the Multiprocess Theory (Einstein et al., 2005; McDaniel & Einstein, 2000), when the delays between the formation of the intention and the opportunity to perform the action are substantial, it would be highly adaptive to have a system that allows spontaneous retrieval so as not to compromise performance of ongoing activities. Thus, in an activity-based task, intention rehearsal during the ongoing task should become almost irrelevant. If so, response times in the ongoing task should not show the typical increase that is observed when people instantiate monitoring processes (Smith, 2008).

Notably, if social goals are sources of activation capable of sustaining activation even without rehearsal, then any effects of goal importance (or reward) on performance accuracy in the activity-based prospective memory task should reflect pre-cognitive, motivation-based mechanisms rather than attention-based mechanisms (cf. Kliegel et al., 2001, 2004) and should be associated with no RT costs or even with a decrease of RTs (Goschke & Kuhl, 1993) in the correspondent conditions of the ongoing task.

2.1. Method

2.1.1. Participants and design

One hundred undergraduates from Suor Orsola Benincasa University participated in the experiment. They were randomly assigned to five between-participants conditions. Namely, standard prospective memory, personal benefit (material reward for the participant), social benefit (benefit for the experimenter/no reward for the participant), social and personal benefits (benefit for the experimenter and material reward for the participant), no prospective memory (ongoing task only).

2.1.2. Procedure and stimulus materials

On arrival, the participant was informed that he/she would perform a computer-based verb identification task. There were 288 three- or four-syllables verbs, half of which were regular and half irregular. Each group of regular and irregular verbs was further divided into three subgroups of 48 verbs each, according to their ending (in Italian, ARE–ERE–IRE for the regular verbs and ARE/RRE, ERE, IRE for the irregular verbs). There were three blocks, each consisting of 48 regular and 48 irregular verbs. Within each block, the 96 items were presented twice, in random order. Overall, each participant provided 576 responses.

The task started with a fixation point that remained on the screen for 500 ms. Then, each verb appeared and remained in view for 5 s. The participants had to press one of two counterbalanced keys, according to whether the

verb was regular or irregular and, at the end of each block, they had to remember to sign a form that was positioned on a table out of the participant's sight. Participants assigned to the no prospective memory condition performed only the verb identification task.

The perceived relevance of the to-be-performed action was manipulated by attaching to the goal different benefits that the action would produce. To this aim, participants received different instructions, according to the experimental conditions. Namely, in the standard prospective memory condition, participants were informed that at the end of each block of verbs they should remember to sign the form. In the remaining conditions, they were additionally informed that if they will remember to sign the form: (a) they will receive one course credit¹ (personal benefit), (b) the experimenter (a Ph.D. student) will collect important data for her thesis (social benefit), or (c) the experimenter will collect important data for her thesis and they will gain one course credit (social and personal benefit). Before starting with the proper experimental session, as well as at the end of it, each participant was required to recall task instructions, to ensure, respectively, that (a) he/she had understood how to do the task and (b) any failure to perform the prospective memory task was not due to forgetting of the content of the to-be-performed action (the *what* and *when* retrospective components).

An additional group of 250 students (group 1) rated, on a seven-point scale, the importance of each of the following everyday life situations: (1) I perform an action and get a small personal benefit (1 course credit); (2) I perform an action that provides a benefit to another person; (3) I perform an action that provides a benefit to another person and I get a small personal benefit (1 course credit). These instructions required the participants to evaluate how important each of the three situations was to them in everyday life and provide a judgment of perceived importance. However, evaluating perceived importance may differ substantially from predicting how likely is that the person will perform the action under different circumstances. Therefore, 62 new participants (group 2) were asked to predict, on a seven-point scale, how likely they would be to perform each of the three actions (see above).

3. Results

All participants recalled correctly *what* to do and *when* they were supposed to do the action. Post-experimental interviews revealed that all participants who *did not sign* the form claimed that they forgot to do the action and showed regret for their failure.

¹ One credit corresponded to about 12.5% of the whole exam. Who got the credit by participating in the experiment earned 12.5% course exemption. Importantly, no participant knew in advance that he/she might be rewarded (none of them would therefore expect to gain a reward). Experimental participants learned about rewards only at the moment of the test. Thus, those who were included in the reward conditions learned about the PM performance-based reward by instructions. Those who were included in the standard PM conditions, received instructions that included no mention to any reward for their PM action.

3.1. Prospective memory accuracy

To analyze prospective memory performance, we computed a 2 (presence/absence of pro-social instructions) $\times$ 2 (presence/absence of reward) ANOVA on the proportion correct values of the four conditions with prospective memory. There was a significant effect of reward, $F(1, 76) = 4.14$, $p < .05$, partial $\eta^2 = .05$, with performance in the no reward conditions being higher than in the reward conditions. However, this effect was modulated by a significant interaction showing that the negative effect of material rewards emerged only under pro-social conditions, $F(1, 76) = 4.14$, $p < .05$, partial $\eta^2 = .05$.

Planned comparisons revealed that participants were more accurate under the pro-social condition, in the absence of material rewards, than under standard prospective memory conditions, $F(1, 58) = 4.47$, $p < .04$, partial $\eta^2 = .07$. Consistent with this result, the number of participants who never signed the form was significantly higher in the standard prospective memory condition than in the pro-social/no reward condition ($\chi^2(40, 1) = 5.71$, $p < .02$). However, when a material reward was introduced under pro-social conditions, it significantly impaired pro-social prospective memory, $F(1, 38) = 8.28$, $p < .006$, partial $\eta^2 = .18$ (Fig. 1). Thus, adding a reward reduced pro-social PM performance to the same level as performance in the standard conditions ($p > .05$).

3.2. Ratings

The resolution of the conflict between competing motives may be either consciously or unconsciously driven. If people are aware of the potential conflict between pro-social and self-gain motives, it seems reasonable to predict that when queried about the importance they attribute to a specific action, they should say that helping others without being rewarded for that is *more* important than helping others and receiving a reward.

To date, when requested to evaluate how important they judged the three situations (the same as those used in the experimental conditions), independent participants

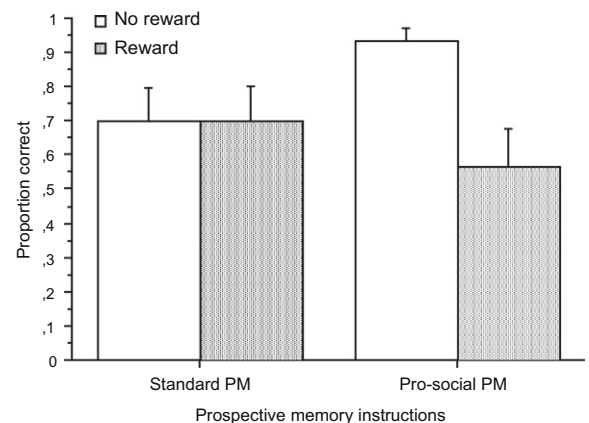


Fig. 1. Prospective memory accuracy as a function of instructions and presence of rewards.

evaluated performing an action and being rewarded for that as less important than helping others without receiving any reward, $F(1, 248) = 34.52$, $p < .0001$, partial $\eta^2 = .12$. However, they evaluated helping others without receiving any reward as less important than helping others and receiving a reward, $F(1, 248) = 5.70$, $p < .02$, partial $\eta^2 = .02$. Importantly, the same pattern emerges when people are asked to *predict* their behavior, rather than to *evaluate* task importance, in the three situations. Namely, participants predicted they would be more likely to perform the action if the action produced a social benefit rather than a personal benefit $F(1, 60) = 17.93$, $p < .0001$, partial $\eta^2 = .23$, and more likely to perform the action if the action produced a social and a personal benefit $F(1, 60) = 8.47$, $p < .004$, partial $\eta^2 = .12$ rather than solely a social benefit. This result argues against the hypothesis that the decrease observed in pro-social PM behavior when a reward is introduced is the by-product of a conscious decision.

3.3. Ongoing task

3.3.1. Response times

We first tabulated the mean response time on correctly answered trials over the three blocks (Table 1). A $5 \times 3 \times 2$ mixed ANOVA that contained the between-subjects variable of prospective memory instructions (no prospective memory, standard, personal benefit, social benefit, personal + Social benefit) and the within-subjects variables of block (One, Two, Three) and Category (regular and irregular verbs) was computed. There was a main effect of Category as participants were slower at identifying irregular than regular verbs, $F(1, 91) = 15.78$, $p < .0001$, partial $\eta^2 = .15$, an effect of block, $F(2, 182) = 34.95$, $p < .0001$, partial $\eta^2 = .28$, indicating that participants speeded up their responses through blocks, and an interaction between Block and Category such that Block improvement was evident for regular but not for irregular verbs, $F(2, 182) = 14.58$, $p < .0001$, partial $\eta^2 = .14$.

To examine the effects of prospective memory instructions, we compared ongoing performance (a) under standard vs. No PM conditions and (b) under standard vs. pro-social conditions. Planned comparisons showed no ongoing costs of performing a standard prospective memory task ($p > .05$). Interestingly, performing a pro-social prospective memory task was associated with faster ongoing task performance, $F(1, 38) = 4.94$, $p < .03$, partial $\eta^2 = .12$. Similarly, mean ongoing performance was faster under social benefit (2575.17 ms) than under standard PM conditions (2829.67 ms), although the effect just failed to reach significance, $F(1, 38) = 3.04$, $p = .08$, partial

Table 2

Percent accuracy in the ongoing (categorization) task.

Condition	Mean proportion	SD
No. PM (control)	.628	.266
Standard PM	.610	.256
Personal benefit	.602	.271
Social benefit	.542	.333
Pers. + Social	.562	.277

$\eta^2 = .07$. Importantly, there was no significant correlation between speed of responding and PM performance across conditions (all $ps > .05$)².

3.3.2. Accuracy

We then computed the proportion of verbs correctly identified as regular vs. irregular. These data were included in $5 \times 3 \times 2$ mixed ANOVA which showed a significant effect of Category, $F(1, 95) = 94.44$, $p < .0001$, partial $\eta^2 = .50$ and an effect of Block, $F(2, 190) = 19.73$, $p < .0001$, partial $\eta^2 = .17$. An interaction between Block and Category was also found, $F(2, 190) = 22.01$, $p < .0001$, partial $\eta^2 = .19$.

As for RTs, ongoing accuracy was compared under PM conditions vs. No PM condition and under standard vs. pro-social conditions. Planned comparisons showed no effect of performing the prospective memory task on the accuracy of performing the ongoing task in the standard and personal benefit conditions ($Fs < 1$). However, participants were less accurate at performing the ongoing task under pro-social PM conditions than under the no prospective memory condition, $F(1, 38) = 9.59$, $p < .003$, partial $\eta^2 = .09$ (Table 2) and less accurate under the social benefit condition than under standard PM condition ($1, 38$) = 6.08, $p < .02$, partial $\eta^2 = .06$. Again, however, there was no significant correlation between ongoing accuracy and PM performance across conditions (all $ps > .05$).

3.3.3. No responses

Because the ongoing task was highly demanding, attentional costs due to intention rehearsal could lead to a higher proportion of no responses in the presence of prospective memory instructions. However, planned comparisons revealed no costs of performing any of the prospective memory tasks as compared to the No PM condition and to the standard PM condition (all $ps > .05$). In contrast, once again, the effects of Category (the rate of no responses was higher for the irregular than for the regular verbs), $F(1, 95) = 94.18$, $p < .0001$, partial $\eta^2 = .50$, and Block, $F(2, 190) = 66.36$, $p < .0001$, partial $\eta^2 = .41$, were significant.

4. Discussion

In this article, we integrated ideas from cognitive, motivational, and social studies to explore the mechanisms

Table 1

Response times in the ongoing (categorization) task.

Condition	Mean	SD
No. PM (control)	2899.65	476.96
Standard PM	2829.67	467.89
Personal benefit	2737.55	566.63
Social benefit	2575.17	584.30
Pers. + Social	2690.38	637.38

² We should, however, recall that mean PM performance in the pro-social condition was almost at ceiling (2.8 out of 3.0). This could make the lack of correlation difficult to interpret. However, in the pro-social/reward condition, where PM performance was far from ceiling (mean: 1.7), there was no correlation, as in the pro-social condition.

underlying pro-social prospective memory. The results indicate that motivation-based mechanisms are major determinants of pro-social prospective memory and posit some constraints to current PM theories, which acknowledge the important role of motivation but, in fact, provide no theoretical advance on the motivational mechanisms of memory for intentions.

Crucially, we showed that altering motivation through the use of rewards affected the level of activation of memory for the pro-social intention. Social incentives (e.g., the goal's social value) improve memory for the intention (Meacham & Kushner, 1980). However, introducing a small material reward under pro-social conditions dramatically decreases memory for the intention and this conflict appears to operate outside of the person's awareness.

Successful performance of an activity-based PM task was accompanied by evidence of no slowing of a demanding ongoing task. On the other hand, in the pro-social conditions there was a reduction of ongoing accuracy. This finding might indicate ongoing costs under pro-social conditions. However, in the conditions under which accuracy decreased, there was a correspondent decrease in RTs. Most important, these ongoing costs were unrelated to PM performance. Indeed, if the decrease in accuracy in the pro-social prospective memory conditions was due to the fact that participants have directed resources away from the ongoing task and towards the prospective memory task, a negative correlation should have emerged between accuracy in the ongoing task and PM performance. However, this was not the case in our study. Although this issue requires further investigation, it is plausible that the effect of introducing a pro-social element to the prospective memory task is to raise the motivational activation of completing the task at hand (i.e., the ongoing task) because instructions for an activity-based task imply "doing something at the end of an ongoing task". This motivation to proceed towards the end may represent an important sub-goal to be fulfilled in order to achieve the final goal (signing). Thus, people speeded up their responses and, as a consequence, they were less accurate. However, shorter latencies and accuracy reduction were unrelated to PM accuracy in the correspondent pro-social conditions (with and without reward), hence making it unlikely that the effect of the pro-social instructions was to raise the importance of the prospective memory task at the expense of the ongoing task. Indeed, if so, better PM performance should have been associated with slower response time. Analogously, worse PM performance should have been associated with higher ongoing accuracy.

In contrast, in this article several positive data support the argument that unconscious motives may prompt forgetting of the intention (under pro-social/reward conditions). Namely: (i) ongoing task RTs decreased under pro-social conditions and this decrease was not associated with PM performance; (ii) there were no costs of performing the prospective memory task on the number of no responses to the ongoing task; (iii) there was a dissociation between people's conscious evaluation of action importance and actual performance; and (iv) there was a dissociation between people's conscious predictions and actual performance.

It appears that, in everyday life, people may not be aware of the possible collision effects of competing motives. To date, the two groups of independent participants not only evaluated helping others without being rewarded for that as less personally important than helping others and gaining a small personal benefit (group 1), but, even more remarkable, they also predicted (Wilson & Gilbert, 2005) they would be less likely to perform the action if they were to help others without reward than with reward (group 2). However, in contrast with the ratings, performance of experimental participants in the pro-social prospective memory conditions was impaired when a material reward was expected. Thus, the competition between pro-social and self-gain motives leads to poor memory for the intention, while remaining largely invisible to the person.

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